Internet of Things (IoT)

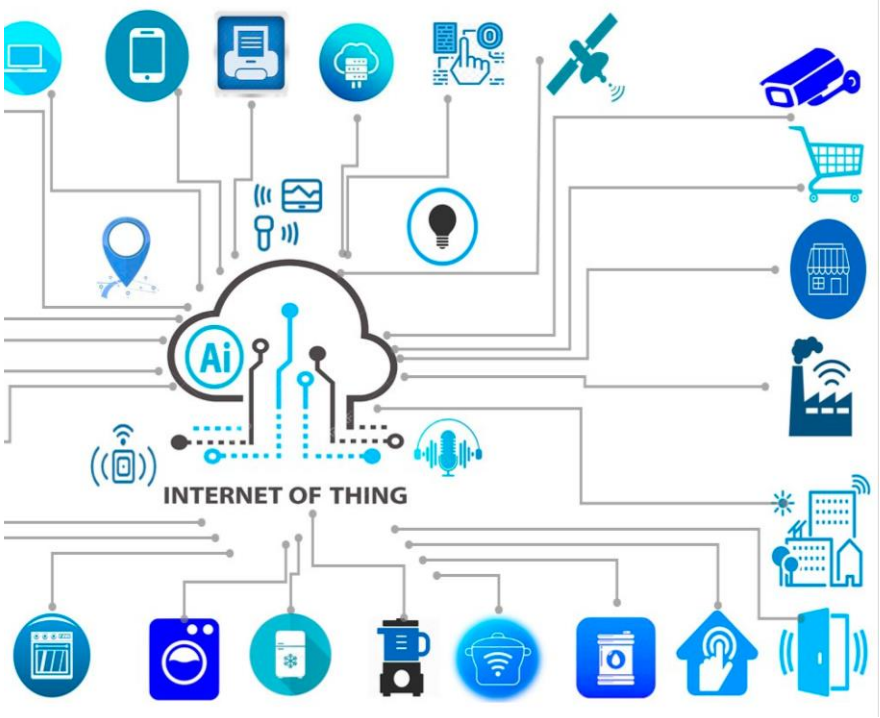
Lecture 2: IoT Architecture & Network Components

Dr. Alyaa Hamza
Alyaa.hamza@sut.edu.eg

Dr. Nehal Anes
nehal.anees@sut.edu.eg

IoT Course Topics

1. Introduction to IoT
2. IoT Architecture & Network Components
3. Smart Objects & IoT Devices
4. Communication Technologies in IoT
5. IoT Protocol Stacks
6. IoT Application Protocols & Data Formats
7. IoT Security Principles
8. Cloud & Fog Computing in IoT
9. Data Analytics in IoT
10. IoT Platforms & Middleware
11. Designing IoT Solutions

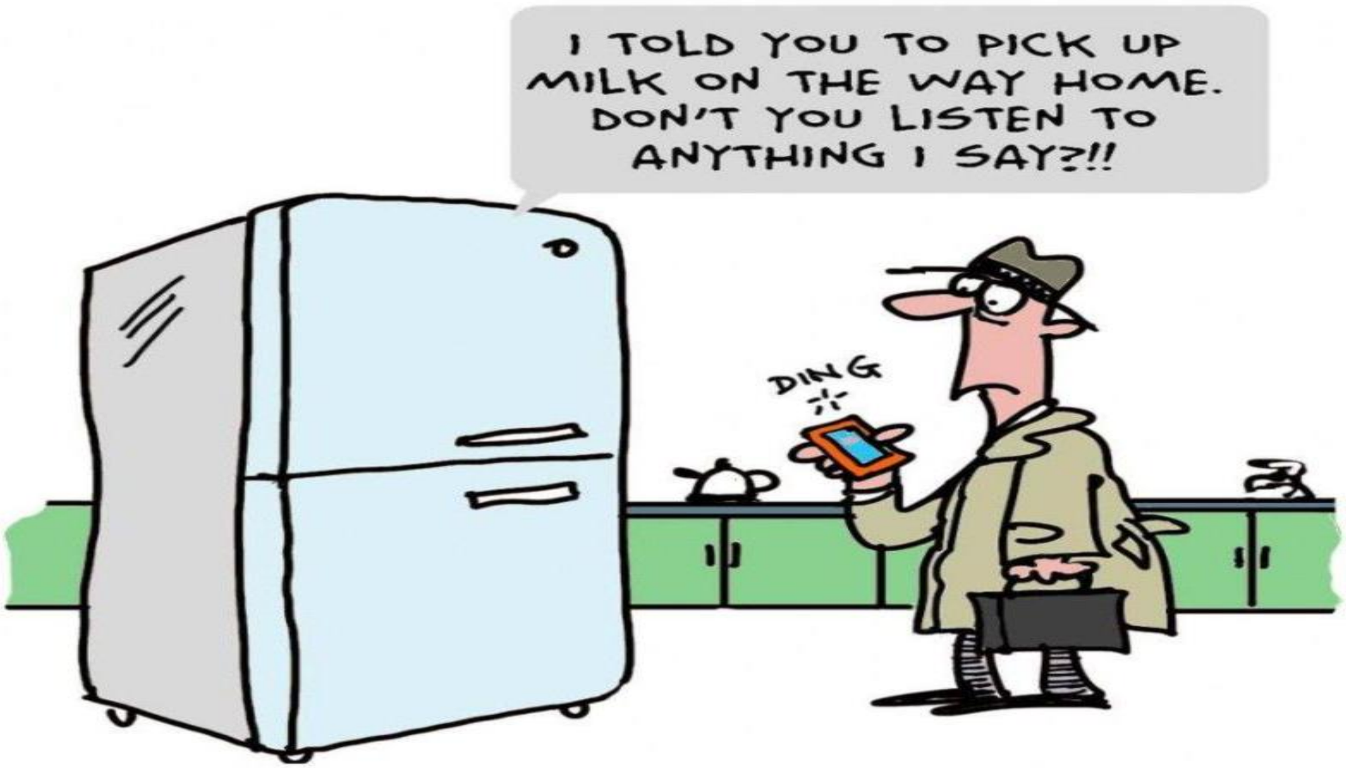


IoT Architecture & Network Components

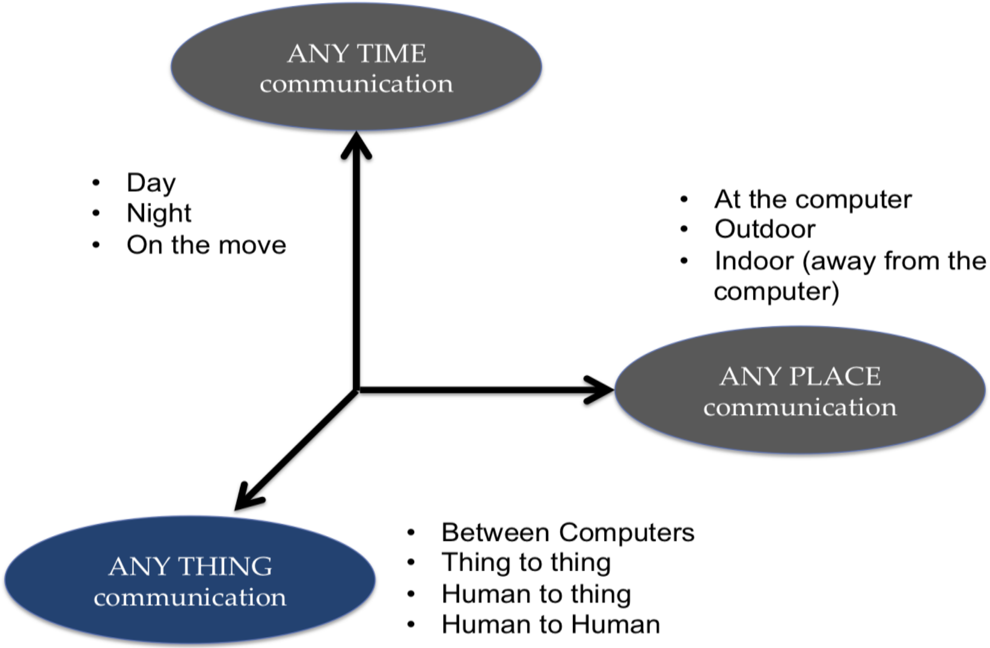
Agenda

1. Building An Architecture
2. Design Principles
3. An IoT Architecture

THE INTERNET OF EVERYTHING



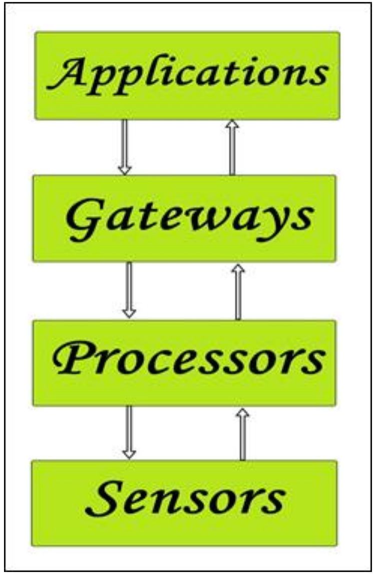
IoT


ANY TIME communication
ANY PLACE communication
ANY THING communication
• Day
• Night
• On the move
• At the computer
• Outdoor
• Indoor (away from the computer)
• Between Computers
• Thing to thing
• Human to thing
• Human to Human


"The Internet of Things
has the potential to
change the world,
just as the internet did,
maybe even more so."
-Kevin Ashton,
Father of IoT

BUILDING BLOCKS of IoT

1) Sensors/Devices
2) Processors
3) Gate Ways
4)Applications



1) Sensors/Devices

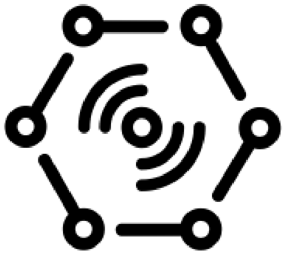
- Sensors or devices are a key component that helps you to collect live data from the surrounding environment.
- These have to be uniquely identifiable devices with a unique **IP address** so that they can be easily identifiable over a large network.
- These have to be **active** in nature which means that they should be able to collect real-time data. These can either work on their own (autonomous in nature) or can be made to work by the user depending on their needs (user-controlled).
- Examples of sensors are gas sensor, water quality sensor, moisture sensor, etc.

2) Processors:

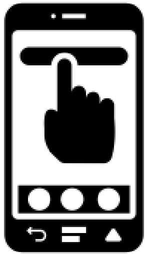
- Processors are the **brain of the IoT system**. Their main function is to process the data captured by the sensors and process them so as to extract the valuable data from the enormous amount of raw data collected.
- Processors mostly work on real-time basis and can be easily controlled by applications. These are also responsible for **securing the data** – that is performing encryption and decryption of data.
- Embedded hardware devices, **microcontroller**, etc are the ones that process the data because they have processors attached to it.

3) Gateways

- Gateways are responsible for routing the processed data and send it to proper locations for its (data) proper utilization.
- Network connectivity is essential for any IoT system to communicate.
- LAN, WAN, PAN, etc are examples of network gateways.



Applications



Applications form another end of an IoT system. Applications are essential for proper utilization of all the data collected.

These cloud-based applications which are responsible for rendering the effective meaning to the data collected.

Applications are controlled by users and are a delivery point of particular services.

Examples of applications are **home automation apps, security systems, industrial control hub, etc.**

Architectural Layers of IoT

- There is **no single** consensus on architecture for IoT, which is agreed universally.

- The most basic architecture is a THREE LAYER defines the main idea of the IoT

Application Layer

Network Layer

Perception Layer

- Researchers often focuses on finer aspects of the IoT. Another proposed architecture is the FIVE LAYER

Business Layer

Application Layer

Processing Layer

Transport Layer

Perception Layer

Building An Architecture

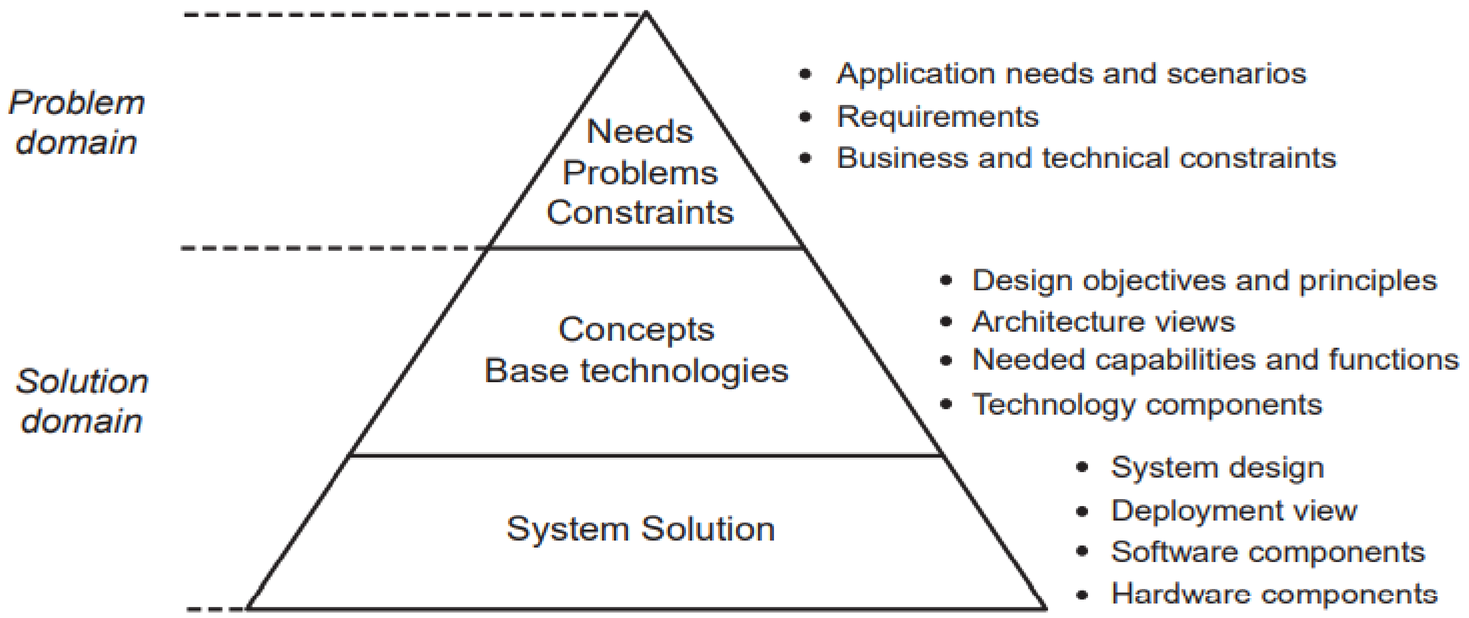
1. When creating a model for the reference architecture

2. one needs to establish overall objectives for the architecture as well as design principles that come from understanding some of the desired major features of the resulting system solution.

3. Problem to solution

4. The problem domain establishes the foundation for the subsequent solutions.

Building An Architecture



Problem and Solution domain partitioning.

DESIGN PRINCIPLES

- Main design principles and needed capabilities (design objective and principles) main desired characteristics of an M2M or IoT solution
- The overall design objective of IoT architecture shall be to target a horizontal system of real-world services that are open, service-oriented, secure, and offer trust.

Design Principles

1. Design for reuse of deployed IoT resources across application domains.

2. Design for a set of support services that provide open service-oriented capabilities and can be used for application development and execution.

- To access iot resources
- How to publish and discover resources,
- Tools for modeling contextual information
- Information related to the real world entities
- Capabilities that provide different levels of abstracted and complex service
- Data filtering and analytics
- Apis & SDK

Design Principles

3. Design for sensing and actors taking on different roles of providing and using services across different business domains and value chains.

- technologies,
- data and service representation
- service : information and services, & aggregation of information

4. Design for sensing and actors taking on different roles of providing and using services across different business domains and value chains.

- Needs to be provided is a set of mechanisms that ensure security and trust
- Need to be authentication and authorization of access to use services as well as to be able to provide services
- Requirement is the capability to be able to do auditing and to provide accountability so that stakeholders can enforce liability if the need occurs
- fundamental requirement is to ensure interoperability

Design Principles

5. Design for ensuring trust, security, and privacy.

Reliability, availability of services, security and privacy

6. Design for scalability, performance, and effectiveness.

Scalability aspects: large number of devices and amounts of data produced
Performance: Supervisory Control And Data Acquisition (SCADA)

7. Design for evolvability, heterogeneity, and simplicity of integration.
8. Design for simplicity of management.
9. Design for different service delivery models (CLOUD SERVICES)

for instance, connected vehicles, and Software as a Service (SaaS) as a delivery model

10. Design for lifecycle support.

The lifecycle phases are: planning, development, deployment, and execution. Management aspects include deployment efficiency, design time tools, and run-time

An IoT architecture outline

we use IoT as a collective term to include both M2M and IoT.

Functional layers and capabilities of an IoT solution.

An IoT architecture outline

Asset Layer. This layer is, strictly speaking, not providing any functionality within a target solution, but justification for existence for any IoT application

The **Resource Layer** provides the main functional capabilities of sensing, actuation, and embedded identities. Sensors and actuators in various devices that may be smartphones or Wireless Sensor Actuator Networks (WSANs), M2M devices like smart meters.

The purpose of the **Communication Layer** is to provide the means for connectivity between the resources on one end and the different computing infrastructures that host and execute service support logic and application logic on the other end. Different types of networks realize the connectivity, and it is customary to differentiate between the notion of a Local Area Network (LAN) and a Wide Area Network (WAN).

An IoT architecture outline

IoT applications benefit from simplification by relying on support services that perform common and routine tasks. These support services are provided by **the Service Support Layer** and are typically executing in data centers or server farms inside organizations or in a cloud environment.

Where the Resource, Communication, and Service Support layers have concrete realizations in terms of devices and tags, networks and network nodes, and computer servers,

the **Data and Information Layer** provided set of functions that capture knowledge and provide advanced control logic support.
Key concepts here include data and information models and knowledge representation in general, and the focus is on the organization of information.
Knowledge Management Framework (KMF) as a collective term to include data, information, domain-specific knowledge, actionable services descriptions

An IoT architecture outline

The **Application Layer in** turn provides the specific IoT applications: Smart Grid, vehicle tracking, building automation, or participatory sensing (PS)

The final layer in our architecture outline is **the Business Layer**, which focuses on supporting the core business or operations of any enterprise, organization, or individual that is interested in IoT applications.
Customer Relationship Management (CRM), Enterprise Resource Planning (ERP), or other Business Support Systems (BSS).

In addition to the functional layers, three functional groups cross the different layers, namely Management, Security, and IoT Data and Services.

An IoT architecture outline

Management, related to system solution related to its operation, maintenance, administration, and provisioning.

This includes management of devices, communications networks, and the general Information Technology (IT) infrastructure as well as configuration and provisioning data, performance of services delivered, etc.

Security is about protection of the system, its information and services, from external threats or any other harm.

Security measures are usually required across all layers, for instance, providing communication security and information security.

An IoT architecture outline

Data and Service processing can,

Basic event filtering and simpler aggregation,
such as data averaging, can take place in individual sensor nodes in WSANs, contextual metadata such as location and temporal information can be added to sensor readings, and further aggregation can take place higher up in the network topology.

More advanced processing is, for instance, data mining and data analytics that can be done in near real-time.

This functional group thus represents the vertical flow of data into knowledge,

Connected World With IOT

We live in a connected world, not only in a social context, but also in a data context. We live in a sea of connected devices which communicate and share information about and with us.


Question
&
Answer